

Lokesh Kumar

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📍 Raipur, Chhattisgarh

TECHNICAL SKILLS AND INTERESTS

Programming Languages: Python, JavaScript (Node.js), C/C++, SQL, HTML, CSS

Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, REST API Design, SDLC, CI/CD Concepts, Unit Testing & Debugging

AI/ML & LLM Tools: Generative AI, LLM Application Development, Prompt Engineering, RAG Systems, AI Copilots, NLP, Model Deployment

Frameworks & Libraries: FastAPI, Django, Flask, Streamlit, ReactJS, LangChain, HuggingFace, TensorFlow, Scikit-learn, Pandas, NumPy

Integration & Automation: REST APIs, Vector Databases (FAISS), Pipeline Automation, Session Management, File Processing

Databases & Cloud: MySQL, MongoDB, Firebase, AWS Basics

Developer Tools: Git, GitHub, Docker, VS Code, Postman

Soft Skills: Problem Solving, Analytical Thinking, Cross-functional Collaboration, Learning new skills, Adaptability

EXPERIENCE

•AI Engineer Internship

June – August 2025

Apana Time

Online

- Designed, developed, and deployed an AI-powered Data Analyst Bot using FastAPI, Streamlit, and LLMs, following structured SDLC practices including requirements analysis, iterative development, testing, and Dockerized deployment.
- Built and integrated REST API endpoints with typed error handling, input validation, and structured prompting, adhering to coding standards and ensuring low defect rates through systematic unit testing and debugging.
- Implemented client-specific features including SQL Query Generation, contextual RAG pipelines, automated data augmentation, session-based memory handling, and multi-format file upload support (CSV, PDF, images) – delivering all within agreed timelines.
- Leveraged AI-first development practices using LLM-based copilots and prompt engineering to accelerate specification writing, improve code quality, and automate analytical workflows – achieving measurable efficiency gains.
- Collaborated across backend and frontend components using Git for version control, Docker for containerized deployment, and Postman for API validation, maintaining consistent coding and quality standards throughout.

PERSONAL PROJECTS

•Enterprise RAG System

Scalable Retrieval-Augmented Generation system for enterprise document intelligence.

- Engineered modular, OOP-based RAG pipeline using FAISS and HuggingFace embeddings for accurate semantic retrieval across large-scale document datasets.
- Implemented hybrid retrieval with dense vector search and BM25 ranking; added unit testing and integration debugging for reliable performance and reduced defects.
- Applied RBAC with metadata-based filtering for secure multi-tenant document access and compliance-oriented retrieval workflows.
- Built REST APIs using FastAPI with a Streamlit frontend; containerized deployment using Docker and managed version control with Git.
- Technology Used: Python, FastAPI, LangChain, FAISS, HuggingFace, Groq (Llama), Streamlit, Docker.

•House Price Prediction API — MLOps Project

Production-ready ML REST API serving a Random Forest model via FastAPI, fully containerized with Docker.

- Developed end-to-end ML pipeline with model training, joblib serialization, modular OOP architecture, structured logging, and Dockerized deployment using Uvicorn ASGI.
- Built REST endpoints (/predict, /health) with Pydantic v2 validation for strict schema and range enforcement; achieved $R^2 \approx 0.80$ on the California Housing dataset.
- Created pytest-based unit tests for API health, prediction flow, and validation failures; automated CI testing with GitHub Actions.
- Managed version control with Git and deployed reproducible containers using Docker on cloud platforms.
- Technology Used: Python, FastAPI, scikit-learn, Pydantic, Docker, pytest, GitHub Actions, joblib, Uvicorn.

•Driver Drowsiness Detection

Real-time computer vision system for driver safety monitoring and alert triggering.

- Developed a real-time eye and facial landmark detection system using OpenCV and Dlib, applying algorithmic problem-solving to monitor Eye Aspect Ratio (EAR) and trigger audio alerts upon drowsiness detection.
- Technology Used: Python, OpenCV, Dlib, NumPy, Pygame.

EDUCATION

•Bachelor of Technology in Artificial Intelligence and Data Science

2022 – 2026

Chouksey Engineering College, Bilaspur

CGPA: 8.55

ACHIEVEMENTS

•4th Place – AlgoVision Data Analytics Hackathon | KOED x IIT Roorkee

2025

- Competed in a national-level data analytics and machine learning competition against participants from multiple institutes; developed data-driven analytical solutions and secured 4th position through effective problem-solving, model development, and collaborative execution.